

UNIT 8

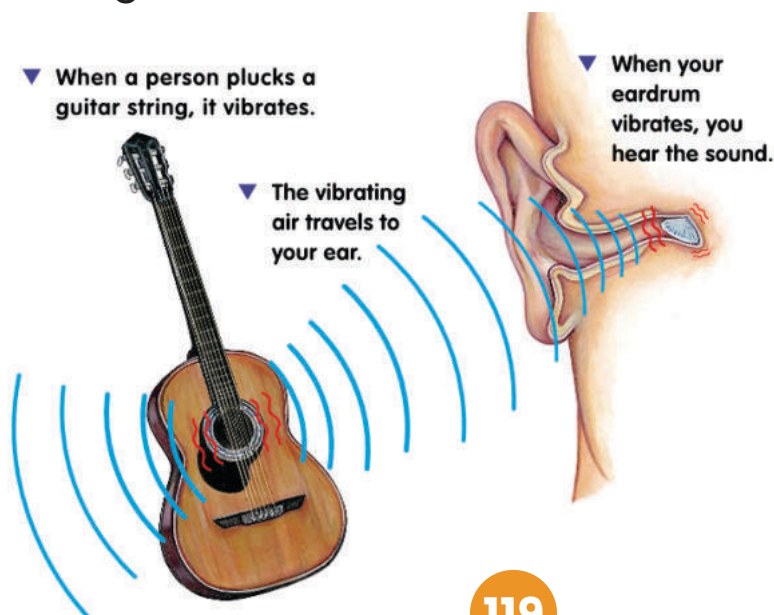
SOUND

Learning objectives:

- Define sound and explain how it is produced.
- Describe how vibrations cause sound.
- Explain how sound travels through different mediums (air, water, solids).
- Differentiate between loud and soft sounds.
- Differentiate between high and low sounds.
- Identify parts of the ear involved in hearing sounds.
- Understand how distance affects the loudness of sound.

What is Sound?

Sound is a kind of energy you hear. Sound is made when something **vibrates**, or moves back and forth.



When you speak, air moves from your lungs to your throat.

The air moves over your vocal cords and causes them to vibrate. Touch your throat as you speak. What do you feel?



◀ When you blow air through a flute, it vibrates and makes a sound.

What Makes Sounds Loud or Soft?

When a sound is loud, the vibrations are big. When a sound is soft, the vibrations are small. When you yell, you make big vibrations in your throat. When you whisper, you make smaller vibrations.



A cat's meow is not as loud as a lion's roar.



A lion's roar is so loud that you can hear it up to 6 miles away.



Think about a lion's roar. When you are near a lion, its roar sounds loud. When you are far away, its roar sounds soft. The farther away you are from a sound, the softer it sounds.



- **What Makes Sounds High or Low?**

Sounds can be high, low, or in between. When a sound is low, the vibrations are slow. A cow's moo and a man's voice are examples of low sounds



▲ When you hit a short bar, you make a high sound.
When you hit a long bar, you make a low sound.

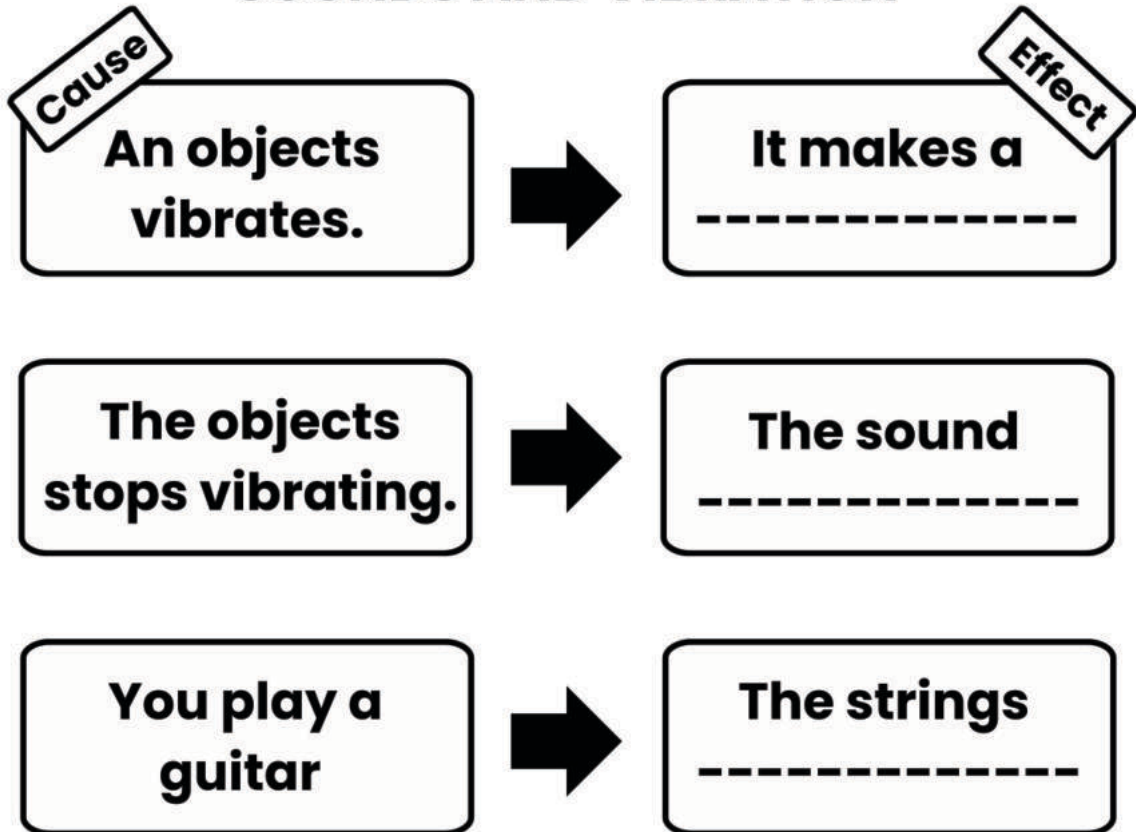


When a sound is high, the vibrations are fast. A cat's meow and a child's voice are examples of high sounds.

Which dog's bark is a higher sound? ▶



SOUNDS AND VIBRATION



Sounds can be loud or soft. Draw something that makes a loud and something that makes a soft sound

Two empty rectangular boxes are provided for drawing. The left box is labeled 'Loud' and the right box is labeled 'Soft'.



Test Your Understanding

Q1: What is sound and how is it produced?

Ans: _____

Q2: How do vibrations create sound?

Ans: _____

Q3: How can you make a sound louder or softer?

Ans: _____

Q4: What part of the ear vibrates to help us hear sounds?

Ans: _____



Q5: What happens to sound when you move farther from its source?

Ans: _____



Fill in the blanks.

1. Sound is a form of _____ that we can hear.
2. When something _____, it produces sound.
3. The part of your ear that vibrates is called the _____.
4. A lion's roar is an example of a _____ sound.
5. A cat's meow is not as _____ as a lion's roar.
6. When you whisper, you create _____ vibrations.



Tick (✓) for True and cross (x) for False statements.

1. Vibrations are fast when a sound is low. ()

2. When you yell, you create bigger vibrations ()

than when you whisper.

3. A child's voice usually makes a high sound. ()

4. The farther you are from a sound, the louder ()

it sounds.

5. When a sound is soft, the vibrations are small. ()



Choose the correct answer:

1. What causes sound?

- a) Light
- b) Vibrations
- c) Heat
- d) Water

2. Which part of the ear helps you hear sound?

- a) Nose
- b) Eardrum
- c) Tongue
- d) Eyes

3. A loud sound has:

- a) Small vibrations
- b) No vibrations
- c) Big vibrations
- d) Slow vibrations

4. Which of the following is a high sound?

- a) A cow's moo
- b) A man's voice
- c) A cat's meow
- d) A lion's roar



5. What happens when you hit a short bar on a xylophone?

- a) It makes a low sound
- b) It makes no sound
- c) It makes a high sound
- d) It makes a soft sound





Activities



ACTIVITY

1.Vibration Experiment: Stretch a rubber band over a box and pluck it. Observe how it vibrates and produces sound. Try plucking harder and softer to see the difference in loudness.

2.Sound Walk: Take a walk around your school or home and list 5 loud sounds and 5 soft sounds you hear. Discuss why some sounds are louder than others.

3.DIY Telephone: Use two paper cups and a string to create a simple telephone. Speak into one cup and listen from the other to understand how sound travels through solids.

